

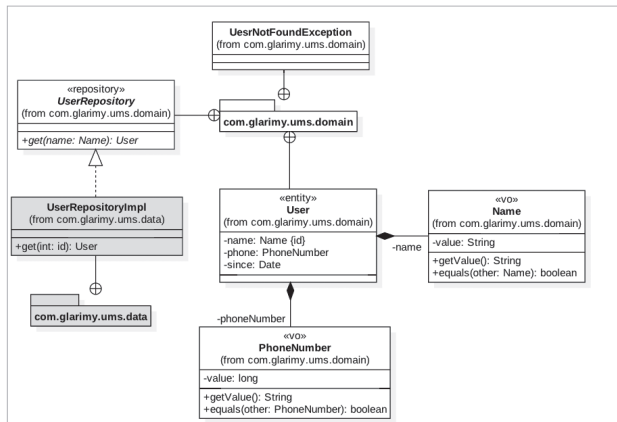


Figure 7: The domain model of FindService

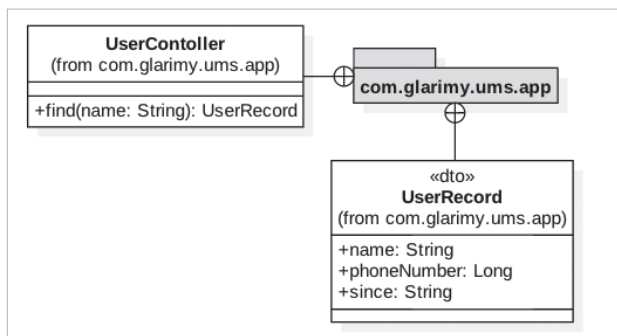


Figure 8: The application layer of FindService

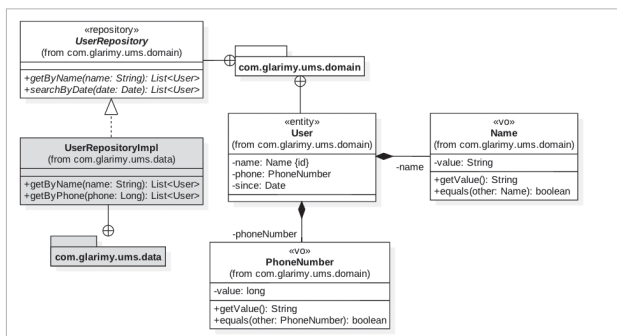


Figure 9: The domain model of SearchService

microservices follows this pattern: 1) Domain model with entities, value objects and repositories, 2) Application layer with controllers and DTOs, and finally, 3) Infrastructure layer with a whole set of generic technical services.

Figure 9 presents the domain model of SearchService and Figure 10 depicts the application layer.

JournalService

And finally, the fourth microservice of the UMS is to listen to the message broker for the events and log them to a centralised journal. Being a separate microservice, this will be developed and deployed independently of the other services.

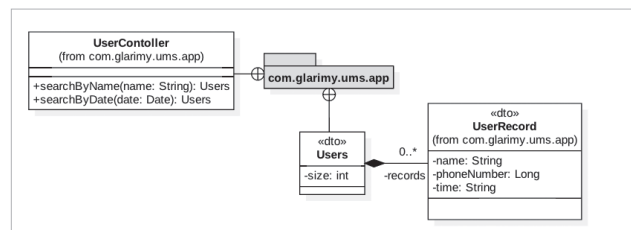


Figure 10: The application layer of SearchService

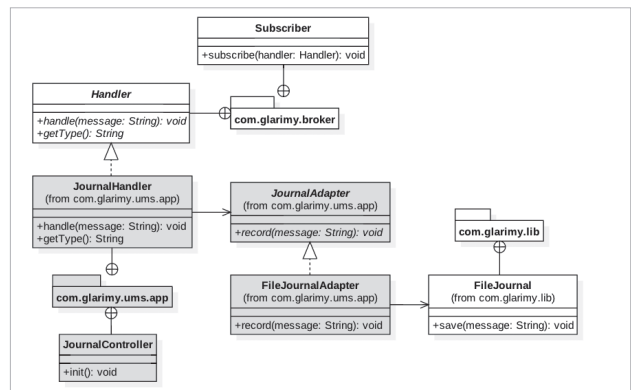


Figure 11: The domain model of JournalService

Refer to the previous parts of this series, for the way we designed journaling in monolith UMS. The design of JournalService has not changed much except for the fact that it is now an independent service. Figure 11 presents the domain model.

Perspective

We have just started our journey into the world of microservices. We have refactored the UMS as a collection of four microservices and presented their models. As mentioned at the beginning of this article, the services may be developed on a common platform or on different platforms. However, they are ultimately deployed on a microservices infrastructure.

What does that infrastructure look like? How about data management across the microservices? How do these services collaborate with each other? How are these services actually developed and deployed? What is the role of Docker? What is the role of Kubernetes?

Well, we will solve these puzzles in the next several parts. **END** 🐧

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